

August 29, 2026

Dear Editor,

I am submitting our manuscript, “Reliable Machine Learning Under Distribution Shift: Uncertainty-Aware Diabetes Risk Prediction from National Health Survey Data,” for consideration as a research article in PLOS Digital Health.

This study examines whether the calibrated probabilities and conformal prediction guarantees used to make machine-learning risk models trustworthy actually hold up under real, non-synthetic distribution shift, using diabetes-risk prediction on the CDC’s Behavioral Risk Factor Surveillance System as the setting. We test four independent shift dimensions, temporal (6-year), demographic subgroup, geographic (US Census region), and model family (deep ensemble vs. gradient-boosted trees), and find that shift magnitude, not merely its presence, determines whether these safeguards hold. Every comparison in the paper carries a 95% bootstrap confidence interval.

We believe this work fits PLOS Digital Health’s scope in machine learning and clinical decision support, particularly its emphasis on rigorous validation using real-world data. The manuscript uses only publicly available, de-identified survey data and involves no human-subjects research requiring IRB approval.

Beyond the specific findings, we believe this work offers the field a template worth reusing: evaluating uncertainty-aware safeguards across several independent, real shift dimensions rather than one, and reporting every comparison with a bootstrap confidence interval rather than a bare point estimate. Both practices are cheap to adopt once a pipeline exists, and our results show that skipping them can hide exactly the failure modes that matter for deployment. We hope this encourages other applied health-ML studies to test claims of “robustness” against more than a single, aggregate shift check.

We confirm this manuscript is not under consideration elsewhere.

Sincerely,

Edmund Eric Gah